# Reduction of the Mathematica result

$\displaystyle\int_{\Lambda/k}^{1-\Lambda/k} d\xi\,\left[\xi\bar\xi+\tfrac{\xi}{\bar\xi}+\tfrac{\bar\xi}{\xi}\right]\log\frac{a}{1-\xi}$ — every step of the algebra

$a=(y-x)^2/m^2$ · the six steps below are exact identities, not expansions

## 0. What is shown

The `ConditionalExpression` is *exactly*, at any $\Lambda/k$,

$$I(a,\lambda)=\mathcal{A}(\lambda)\,\log a+\mathcal{B}(\lambda),$$

$$\mathcal{A}(\lambda)=2\log\frac{1}{\lambda}+2\log(1-\lambda)-\frac{11}{6}+4\lambda-\lambda^2+\frac{2}{3}\lambda^3,$$

$$\mathcal{B}(\lambda)=\frac{1}{36}\Big[18\ell^2-67+36\big(\mathrm{Li}_2(1-\lambda)-\mathrm{Li}_2(\lambda)\big)$$
$$+66\,\omega-18\,\omega^2+138\lambda-12\lambda^2+8\lambda^3+6(\ell-\omega)\big(12\lambda-3\lambda^2+2\lambda^3\big)\Big],$$

with $\lambda=\Lambda/k$, $\ell=\log\frac{1}{\lambda}$, $\omega=\log(1-\lambda)$. Letting $\lambda\to0$ ($\omega\to0$, $\mathrm{Li}_2(\lambda)\to0$, $\mathrm{Li}_2(1-\lambda)\to\frac{\pi^2}{6}$):

$$\boxed{I\;\longrightarrow\;\Big(2\ell_k-\frac{11}{6}\Big)\log\frac{(y-x)^2}{m^2}\;+\;\frac{\ell_k^2}{2}+\frac{\pi^2}{6}-\frac{67}{36},\qquad \ell_k=\log\frac{k^+}{\Lambda}}$$

and what is discarded is $\lambda\big[2\log a+\ell_k\big]+O(\lambda^2\log\lambda)$ — $2.8\times10^{-7}$ at $\lambda=10^{-8}$ (§8).

## 1. Notation

Put $k^+\equiv k$ and

$$\lambda=\frac{\Lambda}{k},\qquad \ell=\log\frac{1}{\lambda}=\log\frac{k}{\Lambda},\qquad \omega=\log(1-\lambda),\qquad A=\log a.$$

The three logarithms in the expression are then

$$\log\frac{\Lambda}{k}=-\ell,\qquad u\equiv\log\frac{ak}{k-\Lambda}=\log\frac{a}{1-\lambda}=A-\omega,\qquad v\equiv\log\frac{ak}{\Lambda}=\log\frac{a}{\lambda}=A+\ell,$$

and the two polynomial brackets, after dividing out $k^3$, are

$$P\equiv\frac{12k^3-12k^2\Lambda+3k\Lambda^2-2\Lambda^3}{k^3}=12-12\lambda+3\lambda^2-2\lambda^3,$$
$$Q\equiv\frac{k^3+12k^2\Lambda-3k\Lambda^2+2\Lambda^3}{k^3}=1+12\lambda-3\lambda^2+2\lambda^3.$$

With these, the whole expression is

$$36\,I=\underbrace{-67+138\lambda-12\lambda^2+8\lambda^3}_{T_0}\;\underbrace{-18u^2+18v^2}_{T_1}\;\underbrace{-6\ell}_{T_2}\;\underbrace{-6u\big(P-6\ell\big)}_{T_3}$$
$$\underbrace{-6\omega}_{T_4}\;\underbrace{+6v\big(Q+6\omega\big)}_{T_5}\;\underbrace{-36\,\mathrm{Li}_2(\lambda)+36\,\mathrm{Li}_2(1-\lambda)}_{T_6}.$$

## 2. Step 1 — the two squared logarithms

Difference of squares, $v-u=\ell+\omega$ and $v+u=2A+\ell-\omega$:

$$T_1=18\big(v^2-u^2\big)=18(v-u)(v+u)=18(\ell+\omega)\big(2A+\ell-\omega\big)$$

$$\Longrightarrow \qquad T_1 = \underline{36A(\ell+\omega)} + 18\big(\ell^2-\omega^2\big).$$

This is where $\ell^2$ enters — $18\ell^2/36 = \ell_k^2/2$ — and where the first half of the $2\ell_k \log a$ comes from.

### 3. Step 2 — the two mixed products; the $\ell\omega$ terms cancel

$$T_3 = -6(A-\omega)(P-6\ell) = -6AP + 36A\ell + 6\omega P - \underline{\underline{36\,\omega\ell}},$$

$$T_5 = +6(A+\ell)(Q+6\omega) = +6AQ + 36A\omega + 6\ell Q + \underline{\underline{36\,\ell\omega}}.$$

The doubly underlined cross terms are equal and opposite and drop out. Adding,

$$T_3 + T_5 = 6A\,(Q-P) + \underline{36A(\ell+\omega)} + 6\,\omega P + 6\,\ell Q.$$

### 4. Step 3 — absorb the two lone logarithms

$T_2 = -6\ell$ and $T_4 = -6\omega$ combine with the last two terms of Step 2:

$$T_2 + 6\ell Q = 6\ell\,(Q-1) = 6\ell\big(12\lambda - 3\lambda^2 + 2\lambda^3\big),$$

$$T_4 + 6\omega P = 6\omega\,(P-1) = 6\omega\big(11 - 12\lambda + 3\lambda^2 - 2\lambda^3\big) = 66\,\omega - 6\omega\big(12\lambda - 3\lambda^2 + 2\lambda^3\big).$$

Both are now manifestly $O(\lambda)$ apart from the single term $66\,\omega$, and they combine into $6(\ell-\omega)\big(12\lambda - 3\lambda^2 + 2\lambda^3\big) + 66\,\omega$ .

### 5. Step 4 — the coefficient of $\log a$

Only Steps 1 and 2 carry $A$. The two underlined pieces add, and

$$P + Q = 13, \qquad Q - P = -11 + 24\lambda - 6\lambda^2 + 4\lambda^3,$$

so the coefficient of $A$ in $36I$ is

$$72(\ell+\omega) + 6(Q-P) = 72\ell + 72\omega - 66 + 144\lambda - 36\lambda^2 + 24\lambda^3,$$

$$\boxed{\mathcal{A}(\lambda) = \frac{1}{36}\Big[72(\ell+\omega) + 6(Q-P)\Big] = 2\ell + 2\omega - \frac{11}{6} + 4\lambda - \lambda^2 + \frac{2}{3}\lambda^3 \xrightarrow[\lambda\to 0]{} 2\ell_k - \frac{11}{6}.}$$

The $-\frac{11}{6}$ is nothing but $\frac{1}{6}(Q-P)\big|_{\lambda=0} = -\frac{11}{6}$: the whole $b$–function coefficient sits in the mismatch between the two polynomial brackets, whose sum is the innocuous 13 and whose difference is $-11$.

### 6. Step 5 — the remainder

Everything not multiplying $A$:

$$36\,\mathcal{B}(\lambda) = \underbrace{18\ell^2}_{\text{Step 1}}\underbrace{-18\omega^2}_{\text{Step 1}}\underbrace{-67 + 138\lambda - 12\lambda^2 + 8\lambda^3}_{T_0}\underbrace{+66\,\omega + 6(\ell-\omega)\big(12\lambda - 3\lambda^2 + 2\lambda^3\big)}_{\text{Step 3}}\underbrace{-36\,\mathrm{Li}_2(\lambda) + 36\,\mathrm{Li}_2(1-\lambda)}_{T_6}.$$

Steps 1–5 are exact rearrangements: $36I = 36\mathcal{A}(\lambda)A + 36\mathcal{B}(\lambda)$ holds as a polynomial identity in $A, \ell, \omega, \lambda, \mathrm{Li}_2(\lambda), \mathrm{Li}_2(1-\lambda)$ (checked with sympy, §9).

## 7. Step 6 — the limit $\Lambda/k \to 0$, piece by piece

| piece of $36\mathcal{B}$ | exact small-$\lambda$ behaviour | $\lambda \to 0$ |
|---|---|---|
| $18\ell^2$ | — | $18\ell_k^2$ |
| $-67 + 138\lambda - 12\lambda^2 + 8\lambda^3$ | $-67 + 138\lambda + \ldots$ | $-67$ |
| $-18\omega^2$ | $-18\lambda^2 + \ldots$ | $0$ |
| $66\,\omega$ | $-66\lambda + \ldots$ | $0$ |
| $6(\ell - \omega)\big(12\lambda - 3\lambda^2 + 2\lambda^3\big)$ | $72\lambda\ell + \ldots$ | $0$ |
| $-36\,\mathrm{Li}_2(\lambda)$ | $-36\lambda - 9\lambda^2 - 4\lambda^3 - \ldots$ | $0$ |
| $+36\,\mathrm{Li}_2(1-\lambda)$ | $6\pi^2 + 36\lambda\log\lambda - 36\lambda + \ldots$ | $6\pi^2$ |
| **total** | | $18\ell_k^2 - 67 + 6\pi^2$ |

(The last two rows use $\mathrm{Li}_2(x) = \sum_{n\geq 1} x^n/n^2$ and $\mathrm{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x) - \mathrm{Li}_2(x)$; the second is needed because $\mathrm{Li}_2$ has a branch point at 1 and `Series` will not expand $\mathrm{Li}_2(1 - \Lambda/k)$ directly.) Dividing by 36,

$$\mathcal{B}(\lambda) \;\longrightarrow\; \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36},$$

and together with $\mathcal{A} \to 2\ell_k - \frac{11}{6}$ from Step 4,

$$\boxed{\int_{\Lambda/k}^{1-\Lambda/k} d\xi\, \frac{P_{gg}(\xi)}{2N_c}\, \log\frac{(y-x)^2}{(1-\xi)\,m^2} = \Big(2\ell_k - \frac{11}{6}\Big)\log\frac{(y-x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} \;+\; O\Big(\frac{\Lambda}{k^+}\Big).}$$

Where each number came from: $2\ell_k$ from the difference of squares (Step 1) plus the mixed products (Step 2); $-\frac{11}{6}$ from $Q - P$ (Step 4); $\frac{\ell_k^2}{2}$ from $18\ell^2/36$ (Step 1); $\frac{\pi^2}{6}$ from $36\,\mathrm{Li}_2(1) = 6\pi^2$; $-\frac{67}{36}$ is the $-67$ that was sitting in the expression from the start.

## 8. What the limit throws away

Collecting the $O(\lambda)$ pieces of $\mathcal{A}$ and $\mathcal{B}$: from $\mathcal{A}$, $2\omega + 4\lambda = 2\lambda + O(\lambda^2)$; from $36\mathcal{B}$, the constants $-72 - 66 + 138 = 0$ cancel and the logarithms combine as $-36\lambda\ell + 72\lambda\ell = 36\lambda\ell$. Hence

$$I(a,\lambda) - I(a,0) = \lambda\Big[2\log a + \ell_k\Big] + O\big(\lambda^2\log\lambda\big), \qquad \lambda = \frac{\Lambda}{k^+}.$$

Verified in §9: the ratio to the measured difference is 0.9999999 at $\lambda = 10^{-8}$. Since $\Lambda/k^+$ is the rapidity cutoff and is taken to zero throughout this framework (cf. the remark preceding Eq. (C.5) of arXiv:1610.03453: "*evaluated under the assumption* $\Lambda \to 0$ *… we will take the limit in all the non–divergent terms without further notification*"), these terms are below the accuracy of everything else in the calculation.

## 9. Checks

Steps 1–5 are verified as exact symbolic identities (sympy), Step 6 numerically.

```
step 1  T1 = 36 A (l+w) + 18(l^2 - w^2)                   : True
step 2  T3+T5 = 6A(Q-P) + 36A(l+w) + 6wP + 6lQ            : True
step 3a T2 + 6lQ = 6l(Q-1) = 6l(12lam-3lam^2+2lam^3)      : True
step 3b T4 + 6wP = 6w(P-1) = 6w(11-12lam+3lam^2-2lam^3)   : True
step 4a P + Q = 13                                        : True
step 4b Q - P = -11 + 24lam - 6lam^2 + 4lam^3             : True
step 4c coefficient of A is  2l + 2w - 11/6 + 4lam - lam^2 + 2lam^3/3 : True
step 5  M/36 = Acoef * A + Brem   (exact identity)        : True
step 6  lambda -> 0 : A*(12*l - 11)/6 + l**2/2 - 67/36 + pi**2/6
        target      : A*(12*l - 11)/6 + l**2/2 - 67/36 + pi**2/6
        equal?      : True
```

```
a = (y-x)^2/m^2 = 100
   lam        direct quadrature     Mathematica verbatim     A(lam)log a + B(lam)         lambda->0 answer
  1e-2       44.496739958518228       44.496739958518228       44.496739958518228       44.359992052876261
  1e-3        78.83843149792572        78.83843149792572        78.83843149792572       78.822329770985848
  1e-4       118.58840748094081       118.58840748094081       118.58840748094081       118.58656559957383
  1e-5       163.65290676920126       163.65290676920126       163.65290676920126       163.65269953864022
  1e-6       214.02075461401265       214.02075461401265       214.02075461401265         214.020731588185
  1e-8       330.66249029501996       330.66249029501996       330.66249029501996       330.66249001870975

what the limit drops:   exact - limit  =  lam [ 2 log a + l_k ] + O(lam^2 log lam)
   lam=1e-2:  exact-limit    0.1367479056   lam(2 log a + l_k)    0.1381551056   ratio 0.989814347
   lam=1e-4:  exact-limit  0.001841881367   lam(2 log a + l_k)  0.001842068074   ratio 0.999898642
   lam=1e-6:  exact-limit  2.302582765e-5   lam(2 log a + l_k)  2.302585093e-5   ratio 0.999998989
   lam=1e-8:  exact-limit  2.763102084e-7   lam(2 log a + l_k)  2.763102112e-7   ratio 0.99999999
```

## 10. Mathematica

```
(* the integral itself *)
lk = Log[k/\[CapitalLambda]];
CUV[xi_] := xi (1 - xi) + xi/(1 - xi) + (1 - xi)/xi;

(* feed Series the inversion identity: Li2 has a branch point at 1 *)
res = res /. PolyLog[2, 1 - x_] :> Pi^2/6 - Log[x] Log[1 - x] - PolyLog[2, x];

Assuming[0 < \[CapitalLambda]/k < 1 && a > 0,
  Normal[Series[res, {\[CapitalLambda], 0, 0}]] // Simplify]

(* -> (2 lk - 11/6) Log[a] + lk^2/2 + Pi^2/6 - 67/36 *)

(* the first correction *)
Assuming[0 < \[CapitalLambda]/k < 1 && a > 0,
  Normal[Series[res, {\[CapitalLambda], 0, 1}]] - % // Simplify]
(* -> (\[CapitalLambda]/k) (2 Log[a] + lk) *)
```